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International AS Level

Physics (9630)
PH01
Report on the Examination

June 2026

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REPORT ON THE EXAMINATION: INTERNATIONAL AS PHYSICS (9630) PH01 – JUNE 2026

This paper performed well, with a mean mark of 49.3.

QUESTION 01

1.1 36% of students gained at least one mark on this question. The main issue was a lack of precision in their explanations. For example, many students talked about “the point” where a change occurred rather stating force, stress, strain or extension.

1.2 This question was more accessible than 1.1. Students were generally clear in their answers, with 54% of students gaining the mark.

QUESTION 02

This question was generally done well; however, some students did not label the particles correctly. The most common error was using “e” or “β” without stating if they were the β^+ or β^-. This meant that they were unable to access the “all symbols correct” mark; this is likely the reason that almost 80% got the first marking point but less than 30% got the second.

QUESTION 03

This question was answered well, with the majority of students gaining full marks.

QUESTION 04

This question discriminated well. 80% of students were awarded the first mark, 63% the second and 44% the third. Few answers were wrong, but many were incomplete or vague.

QUESTION 05

The command word ‘deduce’ generally requires full and clear working, and the wording of the question requires a comparison of their value with 25 N. Some students could not access the third mark due to no comparison or no statement about whether the block would slide or not. Generally, the question was answered well by students.

QUESTION 06

6.1 This was a high-scoring calculation, with 87% of students able to access at least one mark.

6.2 Students found this question challenging. Students struggled to identify the correct laws, and there were many misconceptions. Those who did identify the correct law often had incomplete explanations, so they could not access the mark. Descriptions of Newton’s

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second law were particularly poor. Many students also did not talk about the particles in the question but instead discussed Newton's laws in general terms without linking them to the process in the question.

6.3 This question showed a lot of misconceptions and misunderstandings from students, with only 38% gaining both marks. Many students knew that the two photons were created to conserve momentum but then did not develop that further for the second mark.

QUESTION 07

7.1 This was a simple *suvat* question that was answered well. The only noticeable repeated error in students' answers was confusion of the differing directions of *u* and *g*.

7.2 In this question, students had to specifically mention **kinetic** energy. "Energy" alone was not enough for the mark. 65% of students gained the mark here.

7.3 On the whole, this question was answered well. Some students did not appreciate that they had to find the horizontal component of the velocity before they could use the conservation of momentum equation. This method limited them to 2 of the 4 marks. Over 80% of students accessed at least one mark.

7.4 Some students struggled with power of ten errors on this question or rounded to 1 significant figure for their final answer but, generally, it was answered well.

QUESTION 08

8.1 There were a few routes to the final answer on this question. Most students answered this question well. A frequent issue was rounding too early in the question, leading to an incorrect final answer.

8.2 This was a very accessible question that was answered well, with over 80% of students gaining the mark.

8.3 As is often the case, the extended writing question proved to be a challenge for the students. Over 10% of all students did not even attempt the question. However, over half of students were awarded more than half marks. The students who performed well on this question were those who answered the three bullet points separately and used concise statements and explanations.

QUESTION 09

9.1 This question was answered well, with three-quarters of students gaining full marks. The main errors in this question related to students not understanding components of forces or trying to apply equations without explanation.

9.2 This was a moments question, and over half of students were able to access full marks. Students who did not arrive at the final answer often missed out on method marks due to very poor layout or because they do not clearly present their work.

9.3 This question discriminated quite well. Most students appreciated that the command word **determine** requires a final determination or clear statement that answers the question.

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Some students missed out on the final mark as they did the calculations but did not go on to state whether the latch would engage or not.

QUESTION 10

10.1 The main error for students in this question was confusing reaction time with thinking time.

10.2 In this question, 57% of students gained at least one mark. Most students knew that they should draw a tangent to find the gradient at the steepest part of the line; however, some gradient attempts were very poor.

10.3 When the command word **deduce** is used students are expected to show their working. Most student did this and included annotations on their graphs.

QUESTION 11

11.1 Overall, this question did not score highly. Several students did not add labels or use straight lines from appropriate locations on **S**. Labelled force arrows should always be drawn with a ruler from the point at which they act. Their labels should use standard terms such as weight and tension. The labelling of the downwards force as “G” or “gravity” was a relatively common incorrect answer. Only 17% of students were awarded a mark on this question.

11.2 Generally, this question was well attempted. The majority of students showed a good amount of clear working in their answers.

11.3 Most students used a comparison of the maximum tension and the tension due to the new load; however, there were other comparisons that could be used to gain full marks. Using other routes did lead to some confusion about which way around the comparison should be made, leading to the wrong conclusion.

QUESTION 12

12.1 This was a straightforward question where some students missed marks by using creative ideas for what would cause background radiation. This often led to answers that were too vague to gain the mark.

12.2 Overall, this question was answered poorly. Many students did not appreciate that this is an experiment done by students in schools and therefore gave answers which would not be appropriate in a school context. As with many **explain** questions, some students just stated a technique without explaining the reason for it; this stopped them from accessing the second mark.

12.3 Many students used incorrect significant figures for the calculated value of $\frac{1}{\sqrt{C}}$. As this was a calculated rather than a measured quantity, the significant figures, rather the decimal places, should match the data used. This was the lowest scoring question on the paper with less than 5% of students gaining the mark.

12.4 Lines of best fit should show an even number of points above and below the line. Some students attempted to get as many points on the line as possible; however, this meant that the line was often shifted upwards from where it should be.

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12.5 On the whole, this question was answered well. Most students correctly identified that the line did not go through the origin of the graph. A significant proportion of students also appreciated that the systematic error was due to an underestimate of the distance between the detector and the source.

QUESTIONS 13–26

Section C (multiple choice) was done well, with under 2% not attempting all questions in this section. Whilst students performed well on the majority of questions, Q24 and Q26 were the lowest scoring questions.

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